Boundary CFT$_3$-on-S^3 observables on the truncated GeoVac spectral triple: partition function, wedge KMS entropy, and Casini–Huerta–Myers modular Hamiltonian

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May 2026 (preprint)

Abstract

We exhibit three complementary boundary-side CFT$_3$-on-S^3 observables on the truncated GeoVac spectral triple, each computed from the same wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ via a structurally distinct operation: *(A)* the universal F-theorem coefficient F_{S^3}, extracted via the spectral-zeta derivative $-\frac{1}{2}\zeta'_\Delta(0)$, reproduces the Klebanov–Pufu–Safdi (2011) continuum value bit-exactly for both the conformally coupled scalar ($F_s = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2}$) and the free Camporesi–Higuchi Dirac ($F_D = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2}$); *(B)* the wedge KMS entropy $S(\rho_W) \sim 2\log n_{\max}$ extracts the wedge boundary dimension $\dim(\partial W) = 2$ as a degeneracy log on the lowest K_α^W shell; *(C)* the framework's modular Hamiltonian $K_\alpha^W = J_{\rm polar}$ identifies, at finite cutoff, with the continuum Casini–Huerta–Myers hemisphere modular Hamiltonian, with GH-convergence rate inherited from [8] (and the signature-agnostic product-carrier rate of [14]). The scalar/Dirac partition functions decompose orthogonally under the master Mellin engine of [2]: $F_D + 2F_s = \log(2)/2$ isolates the M2 (Seeley–DeWitt) sector, $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ isolates the M3 (half-integer Hurwitz / odd-zeta) sector. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables. The F-theorem values themselves are due to [23] and are not new; the present paper's contributions are (i) their bit-exact reproduction from the discrete truncated spectral triple, (ii) the orthogonal M2/M3 dual-basis decomposition (Theorem 3.3), and (iii) the finite-cutoff identification of the framework's modular Hamiltonian with the CHM hemisphere operator at GH-convergence rate inherited from [8] (signature-agnostic product-carrier rate in [14]). The bulk-side identifications (Ryu–Takayanagi minimum surfaces, JLMS bulk-modular reconstruction, AdS$_4$ / H^4 dual geometry) are blocked at the framework's existing infrastructure: zero AdS / H^4 machinery, and the Wick-rotation $S^3 \to H^3$ supplies no natural discrete subgroup [5].

1 Introduction

The GeoVac spectral triple [1, 6] is a finite-cutoff operator-system truncation of the Camporesi–Higuchi spectral triple [20] on the round S^3. Its modular structure on the hemispheric wedge has been developed across Papers 42–49 [11–18]: the four-witness Wick-rotation theorem closes at the operator-system level (Riemannian, finite cutoff) via the geometric BW-α generator $K_\alpha^W = J_{\rm polar}$ with integer spectrum, and via the Tomita–Takesaki BW-γ modular operator on the GNS Hilbert–Schmidt space; the modular flow $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ closes bit-exactly. The Lorentzian *propinquity* extension to signature $(3, 1)$ is descoped: Paper 45 [14] proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically, closing that route) and the strong-form construction [15] is correspondingly descoped on the natural substrate, so a

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Lorentzian quantum-metric convergence remains an open target. What survives is the norm-resolvent non-compact-carrier closure [16] at the spectral level and Paper 45's signature-agnostic product-carrier rate. The Krein–Mondino–Sämann bridge [17] and the OSLPLS strong-form bridge [18] provide the synthetic-Lorentzian geometric interpretation.

The present paper turns from the math.OA structural content of the modular machinery to its *physics-side observables*. We ask whether the framework's wedge KMS state ρ_W, evaluated on boundary observables corresponding to standard CFT$_3$-on-S^3 calculations, reproduces the known continuum values, and whether the resulting structural pattern exhibits any genuinely new content beyond a numerical match.

Headline results. We establish three complementary boundary-side observables and one structural decomposition theorem:

1. **Partition function match** (Sec. 3): the universal F-theorem coefficient extracted from the framework's spectral-zeta machinery agrees *bit-exactly* with the Klebanov–Pufu–Safdi (2011) [23] continuum values for both the conformally coupled scalar and the free Camporesi–Higuchi Dirac on the round S^3.

2. **Master Mellin engine decomposition** (Theorem 3.3): the scalar and Dirac partition functions decompose into linear combinations of $\log 2$ and $\zeta(3)/\pi^2$ that project orthogonally onto the M2 and M3 sectors of the master Mellin engine [2]. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables.

3. **Wedge KMS entropy structural classification** (Sec. 4): the von Neumann entropy of ρ_W scales as $S(\rho_W) \sim 2 \log n_{\max}$, with the slope $2 = \dim(\partial W)$ identifying the wedge boundary dimension as a degeneracy log on the lowest K^W_α shell. The F-coefficient does *not* appear in $S(\rho_W)$, demonstrating that the framework's spectral-zeta side and wedge KMS state side encode *different* continuum observables on the same modular state.

4. **Casini–Huerta–Myers identification** (Sec. 5): the framework's modular Hamiltonian K^W_α identifies, at finite cutoff, with the continuum CHM hemisphere modular Hamiltonian $K_{\text{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, d\Omega$ [31]; convergence $K^W_\alpha \to K_{\text{cont}}$ as $n_{\max} \to \infty$ is bounded by the Paper 38 GH-convergence rate $\Lambda \le C_3 \cdot \gamma_{n_{\max}}$ with $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$.

Bulk-side scope. Bulk-side identifications — Ryu–Takayanagi minimum-surface entanglement entropy [33], the JLMS [34] bulk-modular reconstruction, and AdS$_4$ / H^4 dual geometry — are blocked at the framework's existing infrastructure: zero AdS$_4$ / H^4 machinery exists, and a previous attempt at the natural Wick continuation $S^3 \to H^3$ [5] found no discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants. We document this scope statement honestly in Sec. 6, with no claim that the bulk side is reachable from the framework's current axiom set.

2 Setup

2.1 The GeoVac spectral triple on S^3

The Fock projection [1, 19] maps the single-particle hydrogen Hamiltonian conformally to the Laplace–Beltrami operator on S^3 via stereographic projection in momentum space. The framework's spectra on the unit S^3 are exact in $\mathbb{Q}$:

- **Scalar Laplacian.** Eigenvalues $\lambda_n^2 = n(n+2)$ with multiplicity $d_n = (n+1)^2$ for $n = 0, 1, 2, \ldots$. After the conformal mass shift $\xi R_{\text{scalar}} = \frac{1}{8} \cdot 6 = \frac{3}{4}$ for the conformally coupled scalar, the spectrum becomes $(n + \frac{1}{2})(n + \frac{3}{2})$ with the same multiplicity.

- **Camporesi–Higuchi Dirac.** Eigenvalues $|\lambda_n| = n + \frac{3}{2}$ with multiplicity $g_n^{\mathrm{Dirac}} = 2(n+1)(n+2)$ for the single (3d) Dirac fermion (the standard S^3 massless Dirac, whose F_D is reproduced below; in odd dimension there is no Weyl projection), $4(n+1)(n+2)$ for the doubled four-component Dirac. Both carry the same Hurwitz-zeta structure under analytic continuation [3, 4].

The truncation at level $n_{\max}$ retains all $n \le n_{\max}$ shells and defines the operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}} \mathcal{A} P_{n_{\max}}$ in the Connes–van Suijlekom framework [21]. Convergence $\mathcal{O}_{n_{\max}} \to \mathcal{O}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance holds at rate $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$ [8]; the rate constant $4/\pi$ is the M1 Hopf-base-measure signature of the master Mellin engine [2].

2.2 Wedge KMS state

The hemispheric wedge W corresponds, in the truncated full-Dirac basis, to the half-space with $m_j > 0$ along the wedge-fixing axis. Per [11] Def 5.1, the BW-α modular Hamiltonian on the wedge is

$$K_\alpha^W = J_{\mathrm{polar}}$$

where J_{polar} is the rotation generator preserving the wedge, with integer eigenvalues $2m_j \in \{1, 3, 5, \ldots, 2n_{\max}-1\}$ and multiplicity $g(2k+1) = (n_{\max}-k)(n_{\max}-k+1)$ for $k = 0, \ldots, n_{\max}-1$. The wedge KMS state at canonical BW normalization is

$$\rho_W = \frac{e^{-K_\alpha^W}}{Z}, \qquad Z = \mathrm{Tr}_{\mathcal{H}_W}\left(e^{-K_\alpha^W}\right).$$

This is the substrate for all three boundary-side observables that follow.

3 Partition function: bit-exact match to KPS

3.1 Continuum reference

For the round unit S^3, the free energies of free CFTs computed via Hurwitz-zeta regularization are [23]:

$$F_s^{\mathrm{KPS}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.063807, \tag{1}$$

$$F_D^{\mathrm{KPS}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.218960, \tag{2}$$

for the conformally coupled scalar and the free massless Dirac fermion (no Weyl projection in odd dimension), respectively. Both values are expressed in terms of the functional determinant identities $F = -\frac{1}{2}\zeta'_\Delta(0)$ (scalar) and $F = \zeta'_{|D|}(0)$ (Dirac), with the analytic continuation of the divergent spectral sum supplied by the standard zeta-function regularization. The transcendental signature involves only $\log 2$, $\zeta(3)$, and π^2.

3.2 Framework's Hurwitz machinery

The framework's spectral zeta on the truncated S^3 admits a closed-form Hurwitz expansion. For the Dirac case, the Dirichlet series can be written exactly via the half-integer Hurwitz identity $\zeta_H(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$:

$$D_{\mathrm{Dirac}}(s) = \sum_{n=0}^{\infty} \frac{g_n^{\mathrm{Dirac}}}{|\lambda_n|^s} = 2\left(2^{s-2} - 1\right)\zeta_R(s-2) - \tfrac{1}{2}\left(2^s - 1\right)\zeta_R(s). \tag{3}$$

This identity is encoded in the production module `geovac/qed_two_loop.py` and is verified symbolically across integer s.

For the conformally coupled scalar, the spectral zeta admits the Hurwitz-asymptotic expansion

$$\zeta_\Delta(s) = \sum_{n=0}^{\infty} \frac{(n+1)^2}{[(n+1/2)(n+3/2)]^s} = \sum_{k\geq 0} g_k(s)\,\zeta_R(2s+2k-2), \tag{4}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ the binomial-like coefficient.

3.3 Bit-exact match: scalar

Differentiating (4) at $s=0$ and using $g_k(0) = \delta_{k,0}$, $g_k'(0) = 1/(k\cdot 4^k)$ for $k \geq 1$, plus the standard identities $\zeta_R'(-2) = -\zeta(3)/(4\pi^2)$, $\zeta_R(0) = -\frac{1}{2}$, $\zeta_R(2k-2)$ expressible in $\pi^{2k-2}\cdot\mathbb{Q}$ for $k\geq 2$:

$$\zeta_\Delta'(0) = 2\zeta_R'(-2) + \sum_{k\geq 1} \frac{\zeta_R(2k-2)}{k\cdot 4^k}. \tag{5}$$

Theorem 3.1 (Bit-exact scalar partition function)**.** *The framework's scalar zeta-derivative satisfies*

$$F_s := -\tfrac{1}{2}\,\zeta_\Delta'(0) = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} = F_s^{KPS}, \tag{6}$$

in bit-exact agreement with (1).

Proof sketch. The asymptotic Hurwitz expansion (5) converges geometrically in k at rate $1/4$ per term; truncation at $k_{\max} = 100$ gives 61+ matching digits at dps = 200 when compared to $-2F_s^{\mathrm{KPS}}$ via `mpmath`. PSLQ at coefficient bound 10^6 and tolerance 10^{-40} finds the integer relation $[8, 2, -3]$:

$$8\cdot\zeta_\Delta'(0) + 2\log 2 - 3\cdot\frac{\zeta(3)}{\pi^2} = 0,$$

which is exactly the claimed identity. See driver `debug/ads_track_a_scalar_partition_function.py` and JSON `debug/data/ads_track_a_scalar_partition_function.json` for full computation. □

3.4 Bit-exact match: Dirac

Differentiating (3) symbolically at $s=0$, using the same Riemann-zeta identities at $s=-2,0$:

$$D_{\text{Dirac}}'(0) = \tfrac{1}{2}\log 2\cdot\zeta_R(-2) + \left(-\tfrac{3}{2}\right)\zeta_R'(-2) - \tfrac{1}{2}\left[\log 2\cdot\zeta_R(0) + 0\right] \tag{7}$$

$$= 0 - \tfrac{3}{2}\cdot\left(-\tfrac{\zeta(3)}{4\pi^2}\right) - \tfrac{1}{2}\cdot\left(-\tfrac{\log 2}{2}\right) = \frac{3\zeta(3)}{8\pi^2} + \frac{\log 2}{4}. \tag{8}$$

Theorem 3.2 (Bit-exact Dirac partition function)**.** *The framework's Dirac zeta-derivative satisfies, as an exact symbolic identity,*

$$F_D := D_{\text{Dirac}}'(0) = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} = F_D^{KPS}, \tag{9}$$

in bit-exact agreement with (2).

Proof. Direct symbolic computation; `sympy.simplify(D'(0) - F_D)` returns 0. See driver `debug/ads_track_a_dirac_partition_function.py`. □

3.5 Master Mellin engine decomposition

The master Mellin engine of [2] §III.7 classifies the transcendental signatures of GeoVac observables via the three sub-mechanisms

$$\mathcal{M}[\mathrm{Tr}(D^k \cdot e^{-tD^2})], \qquad k \in \{0, 1, 2\},$$

labeled M1 (Hopf-base measure, $\sqrt{\pi} \cdot \mathbb{Q}$ ring at $k = 0$), M2 (Seeley–DeWitt heat-kernel, $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$ ring at $k = 2$), and M3 (half-integer Hurwitz / vertex-parity, $\zeta(3)/\pi^2$ and similar odd-zeta-over-pi ratios at $k = 1$).

Theorems 3.1 and 3.2 exhibit F_s and F_D as linear combinations of $\log 2$ (an M2 transcendental, present via $\zeta_R'(0) = -\frac{1}{2}\log(2\pi)$ in the underlying heat-kernel asymptotic) and $\zeta(3)/\pi^2$ (an M3 transcendental, present via $\zeta_R'(-2) = -\zeta(3)/(4\pi^2)$ in the half-integer Hurwitz-zeta derivative).

Theorem 3.3 (Scalar/Dirac M2–M3 orthogonal decomposition)**.** *The free energies F_s and F_D project orthogonally onto the M2 and M3 axes of the master Mellin engine via the dual-basis combinations*

$$F_D + 2\,F_s = \frac{\log 2}{2} \qquad \textit{(isolates M2; all } \zeta(3) \textit{ cancels)}, \tag{10}$$

$$F_D - 2\,F_s = \frac{3\zeta(3)}{4\pi^2} \qquad \textit{(isolates M3; all } \log 2 \textit{ cancels)}. \tag{11}$$

Proof. Substituting (1) and (2):

$$F_D + 2F_s = \left(\tfrac{\log 2}{4} + \tfrac{3\zeta(3)}{8\pi^2}\right) + 2\left(\tfrac{\log 2}{8} - \tfrac{3\zeta(3)}{16\pi^2}\right) = \tfrac{\log 2}{4} + \tfrac{\log 2}{4} = \tfrac{\log 2}{2},$$

and similarly $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$. The two combinations span a 2-dimensional space; the projection coefficients $(1, +2)$ and $(1, -2)$ act as a dual basis for the M2/M3 decomposition on the (scalar, Dirac) plane. □

Remark 3.4 (First verification on non-GeoVac-internal data)**.** Theorem 3.3 is the first verification of the master Mellin engine on observables that are *not* GeoVac-internal. Previous verifications — the case-exhaustion theorem of [6] §VIII, the 208/208 Prediction 1 panel of [7], the modular-residual closed form of Sprint MR-B[1] — all operated on framework-internal quantities. The KPS partition functions (1), (2) are independently-published continuum CFT_3 data, and the framework's spectral-zeta machinery reproduces them with the exact M2/M3 transcendental signature the master Mellin engine predicts. This extends the engine's reach from a structural classification of GeoVac-internal observables to the organization of independently-published CFT_3 data on the round S^3 at finite cutoff — a bit-exact *reproduction* of the KPS values, not an independent prediction of CFT physics.

4 Wedge KMS entropy: state-side observable

4.1 The framework's wedge KMS entropy

The von Neumann entropy of the wedge KMS state ρ_W at canonical BW normalization $\beta = 2\pi$ is computed directly by diagonalizing K_α^W and applying $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ (recomputed from the live BW construction, the slope-2 scaling, the $\coth 1 - \log(2 \sinh 1)$ constant, and the area-law rejection, in `tests/test_paper50_wedge_kms.py`). Sprint BH-Phase0 (2026-05-22) [36] first reported a small-cutoff log-linear fit $S(\rho_W) \approx 1.94 \log n_{\max} + 0.56$ over $n_{\max} \in \{2, \ldots, 7\}$. A subsequent re-analysis (forcing-catalogue Door 2, 2026-06-01, `debug/door2_bw_entropy_reread_m`

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

replaced this fit by the exact $n_{\max} \to \infty$ resummation of the closed-form entropy $S = \log Z + \langle K_\alpha^W \rangle$ (which reproduces the direct diagonalization to $\le 4 \times 10^{-16}$). The leading $n_{\max}^2$ shell degeneracy $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$ forces the slope to be *exactly* 2, and the additive constant has the exact closed form

$$S(\rho_W) = 2\log n_{\max} + \big(\coth 1 - \log(2\sinh 1)\big) + O(1/n_{\max}), \qquad \coth 1 - \log(2\sinh 1) = 0.458448\ldots, \tag{12}$$

the constant verified by PSLQ to residual $< 10^{-80}$ (integer relation $[-1, 1, -1, -1]$ against $\{C_\infty, \coth 1, \log 2, \log\sinh 1\}$). The windowed least-squares slope climbs monotonically $1.94 \to 2.00005$ as the fit window moves to $n_{\max} \in [350, 2000]$; the original 1.94 is a small-$n_{\max}$ artifact of a sequence converging to exactly 2. The slope $2 = \dim(\partial W) = \dim(S^2_{\text{equator}})$ identifies the wedge boundary dimension as a degeneracy log on the lowest K_α^W shell. The additive constant lives in the Boltzmann-base ring (functions of e via $\coth 1$, $\sinh 1$, tied to the integer spacing $\Delta K_\alpha^W = 2$ at $\beta = 2\pi$); it carries no master Mellin engine transcendental, consistent with Prop. 4.1. A PSLQ search against the F-theorem ring $\{1, \log 2, \zeta(3)/\pi^2\}$ at coefficient ceiling 10^8 returns null: the wedge-entropy constant and the F-coefficient lie in disjoint rings, sharpening Prop. 4.1 from "F absent from $S(\rho_W)$" to "the state-side constant admits no integer relation with the spectral-zeta ring up to coefficient ceiling 10^8" (a strong numerical disjointness, not a proof of ring non-membership; the structural argument that S is a log of an integer, hence ring-free, is the independent support).

4.2 Mechanism: degeneracy log on the lowest K_α^W shell

The K_α^W spectrum is bounded above by $2n_{\max} - 1$ at finite cutoff. At BW canonical normalization, the per-state weight at $K_\alpha^W = 1$ (the equator shell) is e^{-1}/Z, while at $K_\alpha^W = 3$ it drops to e^{-3}/Z, a factor of $e^{-2} \approx 0.135$ suppression per shell. The partition function is therefore dominated by the equator shell of multiplicity $n_{\text{eq}} = n_{\max}(n_{\max} + 1) \sim n_{\max}^2$:

$$Z \approx n_{\text{eq}} \cdot e^{-1} \approx 0.368\, n_{\max}^2,$$

giving effective probability $p_{\text{eq}} \approx 1/n_{\text{eq}}$ per equator state, and hence

$$S(\rho_W) \approx -n_{\text{eq}} \cdot p_{\text{eq}} \log p_{\text{eq}} = \log n_{\text{eq}} \approx 2\log n_{\max}.$$

The two asymptotic limits in s-deformation $\rho(s) = e^{-sK_\alpha^W}/Z$, $s \to 0^+$ (maximally mixed on $\mathcal{H}_W$, giving $S \to \log\dim\mathcal{H}_W$) and $s \to \infty$ (maximally mixed on the equator shell, giving $S \to \log n_{\text{eq}}$), are saturated bit-exactly [36].

4.3 Structural decomposition: F is in the spectral-zeta side, not the state side

Proposition 4.1 (F vs $S(\rho_W)$ live in different sectors)**.** *The framework's wedge KMS state ρ_W admits two structurally distinct calculational operations that extract different continuum observables:*

- ***Spectral-zeta side:*** *the analytic continuation $-\frac{1}{2}\zeta'_\Delta(0)$ extracts the universal F-theorem coefficient F_s^{KPS} via Theorem 3.1; similarly for F_D^{KPS} via Theorem 3.2. The transcendental signature is in the master Mellin engine M2/M3 ring $\mathbb{Q} \cdot \log 2 \oplus \mathbb{Q} \cdot \zeta(3)/\pi^2$.*
- ***Wedge KMS state side:*** *the von Neumann entropy $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ extracts the wedge boundary dimension $\dim(\partial W)$ via degeneracy log on the lowest K_α^W shell, with leading scaling $\dim(\partial W) \cdot \log n_{\max}$. The result is ring-free (a logarithm of an integer).*

The two operations act on the same modular state ρ_W but extract categorically different *continuum observables: the universal F-coefficient versus the wedge boundary dimension. The F-coefficient does* not *appear in $S(\rho_W)$.*

This decomposition is consistent with the PSLQ-disjointness finding of Sprint TD Track 5 (2026-05-09) [37] that GeoVac entropies generally live outside the master Mellin engine ring, while spectral-side observables generally live inside it.

4.4 Three blockers for literal Ryu–Takayanagi on the framework's graph

The continuum Ryu–Takayanagi formula [33] identifies the boundary entanglement entropy of a region A with the area of the bulk minimum surface anchored at ∂A, weighted by $1/(4G_N)$. A literal discrete analog on the framework's graph would require: (i) a well-defined notion of graph minimum-cut between spatial sub-regions; (ii) a perfect-tensor structure at graph vertices ensuring isometric bulk-boundary encoding; (iii) a bulk dual geometry.

Proposition 4.2 (Three blockers for literal RT on the framework)**.** *The framework's discrete S^3 graph at finite $n_{\max}$ exhibits three independent structural obstructions to a literal RT formula:*

1. *The Fock graph at level $n_{\max}$ has $\beta_0 = n_{\max}$ connected components (one per ℓ-sector), so the naive graph-cut between spatial sub-regions is identically zero or trivially confined to a single ℓ-sector.*
2. *The Wigner 3j symbols at graph vertices are Clebsch–Gordan projections rather than isometries; the HaPPY tensor-network RT proof [35] requires perfect tensors at network nodes.*
3. *No bulk-dual construction exists in the framework's infrastructure. The natural candidate — Wick continuation $S^3 \to H^3$ — was closed by Sprint RH-B (2026-04-17) [5] as a clean structural dead end: no discrete subgroup $\Gamma \subset SO(3,1)$ is selectable from framework invariants.*

These three blockers are not in tension with the boundary-side results of Sec. 3 or Sec. 5: the framework reproduces boundary observables on the same wedge KMS state, but cannot independently construct a bulk dual against which to compare via RT.

5 Casini–Huerta–Myers identification

5.1 Continuum CHM hemisphere modular Hamiltonian

The Casini–Huerta–Myers formula [31] for the continuum modular Hamiltonian of a hemisphere of S^3 in free CFT_3 is

$$K_{\mathrm{cont}} = 2\pi \int_{H_+} \xi(x) \cdot T^{00}(x)\, d\Omega(x), \tag{13}$$

where ξ is the wedge-preserving Killing vector field of the rotation that fixes the equator S^2_{equator}, and T^{00} is the local energy density. The flow generated by $K_{\mathrm{cont}}/(2\pi)$ is 2π-periodic, reflecting the canonical Bisognano–Wichmann period of the modular flow [32]; the eigenvalues of K_{cont} in the single-particle basis are $2\pi J$ for integer $J = 1, 2, 3, \ldots$.

5.2 Framework identification

The framework's wedge modular Hamiltonian (Paper 42 Def 5.1) is

$$K^W_\alpha = J_{\mathrm{polar}},$$

the rotation generator of the wedge-preserving $SO(2)$ on the truncated Camporesi–Higuchi spectral triple. Its spectrum is

$$\mathrm{spec}(K_\alpha^W) = \{2m_j : m_j = 1/2, 3/2, \ldots, (2n_{\max}-1)/2\} = \{1, 3, 5, \ldots, 2n_{\max}-1\},$$

integer-valued in the canonical $(2\pi)^{-1}$ rescaling, with multiplicity $g(2k+1) = (n_{\max}-k)(n_{\max}-k+1)$. The modular flow $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ closes bit-exactly at every tested $n_{\max}$ [11]. The odd-integer spectrum, the multiplicities, and the 2π closure are verified from the live BW construction in `tests/test_paper50_wedge_kms.py`.

Theorem 5.1 (CHM identification at finite cutoff)**.** *The framework's modular Hamiltonian K_α^W at finite $n_{\max}$ is the operator-system-level analog of the continuum CHM modular Hamiltonian K_{cont} of (13), with:*

- *Both spectra integer-valued in canonical $(2\pi)^{-1}$ units.*
- *Both generate 2π-periodic modular flow.*
- *The framework's odd-integer spectrum $\{1, 3, \ldots, 2n_{\max}-1\}$ corresponds to the half-integer Casimir grading of the wedge-preserving Killing field on the Weyl spinor sector.*
- *Convergence $K_\alpha^W \to K_{\mathrm{cont}}$ as $n_{\max} \to \infty$ holds in the van Suijlekom state-space Gromov–Hausdorff distance at rate*

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le C_3 \cdot \gamma_{n_{\max}}$$

with $C_3 \to 1$ and $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$ inherited from Paper 38 [8] (the signature-agnostic product-carrier rate of [14] extends it to the temporal factor).

Proof sketch. The wedge-preserving Killing field of the round S^3 rotates the spinor harmonics by their m_j eigenvalue; the canonical 2π-period of the rotation gives the integer $2m_j$ spectrum in the canonical normalization. The framework's truncation $K_\alpha^W = J_{\mathrm{polar}}$ acts on the projected wedge sub-space $P_W \mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ with exactly the same $2m_j$ eigenvalues, restricted to the multiplicities allowed by the truncation. Convergence in the state-space Gromov–Hausdorff distance follows from [8]. □

6 Bulk-side scope

The three boundary-side observables (Sec. 3, Sec. 4, Sec. 5) all reproduce or identify their continuum CFT$_3$-on-S^3 analogs at finite cutoff. The corresponding *bulk-side* identifications — AdS$_4$ gravity, JLMS bulk-modular reconstruction, RT minimum surfaces, holographic entanglement entropy — are blocked at the framework's existing infrastructure.

Proposition 6.1 (Bulk side blocked)**.** *None of the bulk-side identifications corresponding to Sec. 3–5 are reachable from the framework's current infrastructure:*

1. *No AdS$_4$ / H^4 machinery exists in the framework's codebase (`geovac/`); a search for AdS, hyperbolic-4-space, or holographic primitives returns no matches.*
2. *Sprint RH-B (2026-04-17) [5] closed the natural candidate — Wick continuation $S^3 \to H^3$ — as a clean structural dead end: the continuation supplies no natural discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants.*
3. *Sprint L3e-P3 (2026-05-23) found that Paper 38's $4/\pi$ rate does* not *transport to the non-compact Coulomb setting, suggesting that any discrete-to-bulk convergence statement would require a non-compact rate analog that is itself an open NCG-math problem.*

The structural reading of Prop. 6.1 is that the bulk-side identification falls into Class 1 (calibration-external) of the framework's external-input partition [38]: the framework determines the boundary-side observables structurally but does not autonomously generate the bulk-dual geometry. Importing AdS_4 infrastructure would be a major structural extension, beyond sprint scale and outside the present paper's scope.

7 Extension to S^5: structural refinement of the dual-basis decomposition

The bit-exact match of Sec. 3 extends to round S^5 via the same Hurwitz-zeta machinery. The Camporesi–Higuchi spectra on S^5 are spectrally analogous to S^3: the conformally coupled scalar has eigenvalues $(n+3/2)(n+5/2)$ with multiplicity $(2n+4)(n+1)(n+2)(n+3)/24$ for $n = 0, 1, 2, \ldots$, and the Weyl Camporesi–Higuchi Dirac has $|\lambda_n| = n + 5/2$ with multiplicity $(n+1)(n+2)(n+3)(n+4)/12$. The framework's spectral-zeta machinery on these spectra produces partition function values in a closed-form ring extending [23]'s S^3 ring by one $\zeta(5)/\pi^4$ generator.

7.1 Bit-exact closed forms on S^5

Theorem 7.1 (Conformally coupled scalar on S^5). *The framework's scalar zeta-derivative on round unit S^5 satisfies*

$$F_s^{S^5} := -\tfrac{1}{2}\,\zeta'_{\Delta_{conf}}(0)^{S^5} = -\frac{\log 2}{128} - \frac{\zeta(3)}{128\pi^2} + \frac{15\,\zeta(5)}{256\pi^4} \approx -5.74 \times 10^{-3}. \tag{14}$$

Proof sketch. The Hurwitz expansion

$$\zeta^{\text{conf}}_{S^5}(s) = \frac{1}{12}\sum_{k\geq 0} g_k(s)[\zeta_R(2s+2k-4) - \zeta_R(2s+2k-2)] \tag{15}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ follows from the re-indexed multiplicity structure $v^2(v^2-1)/12$ at $v = n+2$. At $s = 0$, only $g_0(0) = 1$ survives, giving $\zeta^{\text{conf}}_{S^5}(0) = 0$; the derivative

$$\zeta'_{S^5}(0) = \frac{1}{6}[\zeta'_R(-4) - \zeta'_R(-2)] + \frac{1}{12}\sum_{k\geq 1}\frac{\zeta_R(2k-4) - \zeta_R(2k-2)}{k\cdot 4^k} \tag{16}$$

converges geometrically in k. PSLQ at 200 dps and $k_{\max} = 100$ finds the integer relation $[32, -2, -2, 15]$ (applied to $4\,\zeta'_{S^5}(0)$) bit-exactly identifying

$$\zeta'_{S^5}(0) = \frac{\log 2}{64} + \frac{\zeta(3)}{64\pi^2} - \frac{15\,\zeta(5)}{128\pi^4}. \tag{17}$$

The standard identities $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$ and $\zeta'_R(-4) = +3\zeta(5)/(4\pi^4)$ deliver the $\zeta(5)/\pi^4$ contribution; the $\log 2$ piece arises from $\zeta'_R(0) = -\frac{1}{2}\log(2\pi)$ via the Hurwitz analytic-continuation tail. □

Theorem 7.2 (Weyl Camporesi–Higuchi Dirac on S^5). *The framework's Weyl Camporesi–Higuchi Dirac zeta-derivative on round unit S^5 satisfies, as an exact symbolic identity,*

$$D'_{Dirac}(0)^{S^5} = -\frac{3\log 2}{128} - \frac{5\,\zeta(3)}{128\pi^2} - \frac{15\,\zeta(5)}{256\pi^4} \approx -0.02163. \tag{18}$$

Proof sketch. The Dirichlet series

$$D_{\text{Dirac}}(s)^{S^5} = \frac{1}{12}\left[\zeta_H(s-4, 5/2) - \tfrac{5}{2}\zeta_H(s-2, 5/2) + \tfrac{9}{16}\zeta_H(s, 5/2)\right] \tag{19}$$

follows from re-indexing $u = n + 5/2$ in the multiplicity $(n+1)(n+2)(n+3)(n+4)/12 = (u^2 - 9/4)(u^2 - 1/4)/12$. The half-integer Hurwitz function $\zeta_H(s, 5/2)$ has closed form $(2^s - 1)\zeta_R(s) - 2^s - (2/3)^s$. Differentiating each shift symbolically at $s = 0$ and combining via the multiplicity coefficients $(1, -5/2, 9/16)$ delivers the claimed result. □

7.2 Log 3 cancellation theorem

Theorem 7.3 (log 3 cancellation on S^5)**.** *The $(2/3)^s$ terms in $\zeta_H(s-a, 5/2)$ for $a \in \{0,2,4\}$ contribute, at $s=0$, factors of* $\log(2/3) = \log 2 - \log 3$ *scaled by* $(2/3)^{-a} = (3/2)^a$*. The coefficient of* $\log 3$ *in* $D'_{Dirac}(0)^{S^5}$ *evaluates to*

$$\frac{1}{12}\left[\frac{81}{16} - \frac{5}{2}\cdot\frac{9}{4} + \frac{9}{16}\cdot 1\right] = \frac{1}{12}\left[\frac{81-90+9}{16}\right] = 0 \tag{20}$$

exactly.

Remark 7.4. The structural combination $(1, -5/2, 9/16)$ encoded in the multiplicity polynomial $(u^2-9/4)(u^2-1/4) = u^4-(5/2)u^2+9/16$ aligns the three shift operators $\{\zeta_H(s,5/2), \zeta_H(s-2,5/2), \zeta_H(s-4,5/2)\}$ such that all $\log 3$ contributions cancel exactly. The framework's spectral content on S^5 stays in the M-engine ring $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4\}$, even though the naive Hurwitz expansion suggests a contribution from $\log 3$. The same mechanism extends the M2/M3 ring statement from S^3 to S^5.

7.3 Dual-basis projection: structural-specificity to S^3

Theorem 3.3 of Sec. 3 establishes that on S^3, the (scalar, Dirac) pair admits a dual-basis projection onto the M2/M3 axes: $F_D + 2F_s = \log(2)/2$ (M2 only), $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ (M3 only). On S^5, the analogous statement *fails*.

Proposition 7.5 (Dual-basis non-extension to S^5)**.** *The Paper 50 Theorem 3.3 dual-basis projection does not extend to S^5. The M-engine ring on S^5 is three-dimensional (*$\log 2$*,* $\zeta(3)/\pi^2$*,* $\zeta(5)/\pi^4$*), but the (scalar, Dirac) plane is two-dimensional. The system $aF_s^{S^5} + bF_D^{S^5} = c \cdot$ (single M-axis) is over-determined: imposing isolation of one M-axis requires three linear conditions on two parameters (a,b), which has no nontrivial solution.*

Explicitly, attempting to isolate $\log 2$*:*

$$\zeta(3)/\pi^2 \ \textit{coefficient} = 0 \implies a + 5b = 0 \implies a = -5b, \tag{21}$$

$$\zeta(5)/\pi^4 \ \textit{coefficient} = 0 \implies 15a - 15b = 0 \implies a = b, \tag{22}$$

which are inconsistent unless $b = 0$.

Remark 7.6 (Continuation of the F-coefficient ladder through S^{11}; dual-basis non-extension)**.** Two statements must be distinguished here: the *single-species* in-ring closure of F (which continues to all odd d tested), and the *dual-basis projection* of Theorem 3.3 (which is special to S^3). The structural-specificity of Theorem 3.3 concerns only the second.

Single-species in-ring closure (continues through S^{11}). For each free species individually, $F = -\frac{1}{2}\zeta'_\Delta(0)$ on round S^d (d odd) lies in the ring

$$\mathcal{R}_d = \{\log 2\} \cup \{\zeta(2k+1)/\pi^{2k} : k = 1, \ldots, (d-1)/2\}$$

of dimension $(d-1)/2 + 1$ (top atom $\zeta(d)/\pi^{d-1}$). This is verified bit-exactly at S^3, S^5 (Theorems 7.1, 7.2), and — by direct high-precision computation (Door 1 graduation, 2026-06-03; 260 dps, frozen basis, reconstruction error $\lesssim 10^{-261}$, decoy-controlled) — at S^7, S^9, and S^{11}. The conformally coupled scalar on S^{11}, for instance, is

$$\begin{aligned}\zeta'_{\mathrm{sc},S^{11}}(0) = &-\tfrac{7}{131072}\log 2 - \tfrac{3897}{45875200}\tfrac{\zeta(3)}{\pi^2} - \tfrac{485}{8257536}\tfrac{\zeta(5)}{\pi^4} \\ &+ \tfrac{609}{1310720}\tfrac{\zeta(7)}{\pi^6} + \tfrac{425}{262144}\tfrac{\zeta(9)}{\pi^8} + \tfrac{1023}{524288}\tfrac{\zeta(11)}{\pi^{10}},\end{aligned} \tag{23}$$

and the free Weyl Camporesi–Higuchi Dirac value is symbolic-exact in the same ring. A decoy test assigns Catalan G, $\log 3$, and unscaled $\zeta(3)$ coefficient exactly zero, confirming a clean M3 signature.

Erratum. An earlier version of this remark reported S^7 as a structural non-match ("no PSLQ relation in the tested bases"). That was a *false negative* from an under-resolved search. The genuine S^7 scalar relation

$$\zeta'_{\mathrm{sc},S^7}(0) = -\tfrac{1}{512}\log 2 - \tfrac{41}{15360}\tfrac{\zeta(3)}{\pi^2} + \tfrac{5}{1024}\tfrac{\zeta(5)}{\pi^4} + \tfrac{63}{2048}\tfrac{\zeta(7)}{\pi^6}$$

has integer coefficients of order 10^4, but locking it requires the value to $\gtrsim 200$ digits; a 30-digit value run against maxcoeff $= 10^{16}$ sits below the resolution threshold and returns spurious non-detection. Verified independently at 200 dps (reconstruction error $\approx 2.4 \times 10^{-202}$) and reproduced by the Door 1 driver at 260 dps. The ladder therefore *generates*: every odd rung $S^3, \ldots, S^{11}$ has a rational-coefficient closed form in $\mathcal{R}_d$, with no transcendental outside $\mathcal{R}_d$.

Dual-basis projection (special to S^3, unaffected). Isolating a single M-axis using only the (scalar, Dirac) pair requires $\dim \mathcal{R}_d = 2$, which holds at S^3 alone. For $d \geq 5$ the projection is over-determined (Prop. 7.5); the related cross-species ladder recursion scalar $- m \cdot$ Dirac $= c_d \cdot \mathrm{Dirac}_{S^{d-2}}$ likewise closes at S^5 only, with the top-atom ratio $\mathrm{scalar}_{\mathrm{top}}/\mathrm{Dirac}_{\mathrm{top}} = 2^{-(d-5)/2}$ hitting the required value ($m = 1$) at $d = 5$ alone (Door 1 graduation). Thus the F-coefficients are in-ring at every odd d, but the two-species dual basis that *spans* the ring is special to S^3. Supplying the missing higher-spin free species — $(d-1)/2 - 1$ of them at S^d — to span $\mathcal{R}_d$ remains the natural continuation, and is the actual structural-specificity content of Theorem 3.3.

7.4 Implementation effort and continuation pattern

The S^5 extension was computed in 35 minutes of focused main-session work, continuing the discrete-tractability advantage documented across the math.OA arc. The `debug/ads_track_a_s5_{scalar,dirac}_partition_function.py` scripts implement the Hurwitz expansions and PSLQ verifications; `debug/ads_track_a_s5_extension_memo.md` documents the findings in full.

8 Synthesis and open questions

8.1 Unified picture

The framework's wedge KMS state $\rho_W = e^{-K^W_\alpha}/Z$ supports three structurally distinct calculational operations, each extracting a different continuum CFT$_3$-on-S^3 boundary observable:

Operation	Continuum analog	Ring
$-\frac{1}{2}\zeta'_\Delta(0)$ (Theorem 3.1)	F-theorem coefficient	M2 + M3
$-\mathrm{Tr}[\rho_W \log \rho_W]$ (Prop. 4.1)	Wedge boundary dimension	ring-free
$\mathrm{spec}(K^W_\alpha)$ (Theorem 5.1)	CHM modular Hamiltonian	ring-free

All three converge to their continuum analogs as $n_{\max} \to \infty$ in the van Suijlekom state-space Gromov–Hausdorff distance per [8] (with the signature-agnostic product-carrier rate of [14] on the temporal factor).

8.2 Open questions

1. **Other free-field CFTs.** The scalar and Dirac cases exhibit the orthogonal M2/M3 decomposition of Theorem 3.3. Does the same projection structure extend to higher-spin partition functions on S^3, in particular the Maxwell ($p = 1$ form) and graviton ($p = 2$) cases? The Maxwell partition function on round S^3 has known closed forms [26]; testing whether the framework's spectral-zeta machinery reproduces it (and whether the M2/M3 decomposition extends) is a natural follow-up.

2. **Squashed-S^3 generalization.** The KPS values above are for the round (unit) S^3. The squashed-S^3 partition functions of supersymmetric CFTs [25] use deformed half-integer Hurwitz machinery that may not reduce to the framework's MR-B Jacobi-ϑ_2 closed form. Whether the framework can match the squashed case is open.

3. **State-space-GH-rate convergence statement.** Theorem 5.1 states convergence at the state-space Gromov–Hausdorff level (inherited from Paper 38). A quantitative rate for the partition-function-side convergence $|F_{n_{\max}} - F^{\mathrm{KPS}}| \leq C \cdot \gamma_{n_{\max}}$ with explicit C requires a Mellin-domain version of the Berezin-reconstruction framework of Paper 38; this is open work.

4. **Bulk-side identification (Prop. 6.1).** The JLMS bulk-modular Hamiltonian identification, the RT entanglement-entropy formula, and the AdS$_4$ dual geometry remain open. Importing AdS$_4$ / H^4 infrastructure is the named major structural extension that would reach these; it is beyond sprint scale.

8.3 Catalogue and open targets

The bit-exact matches at S^3 and S^5 exhibit a pattern: the framework's discrete spectral-zeta machinery reproduces *independently-published* continuum CFT-on-sphere free-energy values, and the transcendental signature decomposes via the master Mellin engine [2]. This pattern admits systematic extension. The following table catalogues the current status:

System	Continuum reference	Fra
Free scalar (conformal) on S^3	KPS [23]	D
Free Dirac (CH) on S^3	KPS [23]	D
Free scalar (conformal) on S^5	Beccaria-Tseytlin [26]	D
Free Dirac (CH) on S^5	Beccaria-Tseytlin [26]	D
Free scalar (conformal) on S^7	— (framework-generated)	DONE (in-ring $\mathcal{R}_7$, $\geq$
Free Maxwell on S^3	3D scalar duality [26]	open: S^3 Maxwell-sc
Free Maxwell on S^5	BT [26]	TODO; needs tran
Free higher-rank tensor ($p \geq 2$) on S^d	BT [26], Anninos	TODO; need
Free (non-)minimal scalar (zero mode) on S^d	—	open: m=2 edge correc
Squashed S^3 partition function	JKPS, Hama-Hosomichi-Lee	open: deformed half-integer Hu

Holographic Weyl anomaly $\leftrightarrow$ Connes–Chamseddine spectral action: both live in M2. A structural classification shared with the AdS/CFT literature that the framework obtains directly — without additional computation — is the M2-sector placement of the holographic Weyl anomaly and the Connes–Chamseddine spectral action.

Henningson and Skenderis [28] derived the Weyl anomaly of CFT$_d$ on a manifold M via the bulk AdS$_{d+1}$ gravitational action regularized at a finite radial cutoff. For even d, the result reproduces the boundary central charges $a_{\mathrm{CFT}}, c_{\mathrm{CFT}}$ as specific combinations of bulk gravitational coefficients. The bulk integral is computed via the Riemannian curvature data of AdS, evaluated through a heat-kernel expansion in the AdS-radius cutoff ϵ.

The Connes–Chamseddine spectral action [30] $\mathrm{Tr}\, f(D/\Lambda)$, on any almost-commutative spectral triple, has heat-kernel expansion coefficients given by Seeley–DeWitt coefficients $a_k(D^2)$. For the GeoVac spectral triple on round S^3, the Seeley–DeWitt coefficients of D^2_{CH} are exact in $\sqrt{\pi} \cdot \mathbb{Q}$ with no higher coefficients (Paper 28 [9], Sprint MR-B closed form, two-term-exact theorem). The spectral action therefore lives in the M2 ring of the master Mellin engine (Paper 18 [2] §III.7).

Structural classification. Both the Henningson–Skenderis holographic Weyl anomaly and the Connes–Chamseddine spectral action sit in the master Mellin engine's M2 sub-mechanism (Seeley–DeWitt heat-kernel ring $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$). This is a shared heat-kernel *classification*, not a computed identity: both are heat-kernel-trace observables ($\mathcal{M}[\mathrm{Tr}\, e^{-tD^2}]$ at operator order $k = 2$), the HS quantity computed on the bulk-gravity side and the CC quantity on the finite spectral triple. The framework does not compute the HS holographic anomaly; the content is the taxonomic placement of both in the M2 ring. The framework's discrete-cutoff realization (Paper 38 [8] state-space GH convergence at rate $4/\pi$) is the operator-system-level home of the CC side.

This is the framework's most direct contact point with established holographic computations in the M2 sector and is a corollary of the master Mellin engine case-exhaustion theorem (Paper 32 §VIII), not a new computation. The substantive content is the explicit identification of HS's bulk computation as an M2-sector observable, placed in the same heat-kernel ring as the framework's spectral action on S^3 — a structural classification, not a reproduction of the holographic computation.

Relation to the TT-bar cut-off AdS picture. Hartman, Kruthoff, Shaghoulian, and Tajdini [29] showed that sphere partition functions of TT-bar-deformed large-N holographic CFT_d in $d \in \{2, 3, 4, 5, 6\}$ non-perturbatively match bulk computations of AdS_{d+1} at finite radial cutoff. The parallel to the present paper's discrete-cutoff framework ($\mathcal{T}_{n_{\max}}$ at finite $n_{\max}$ matching continuum CFT) is suggestive, but the two cut-off structures are *categorically distinct*:

- **TT-bar (Hartman et al. 2019):** a *deformation of the action*. The TT-bar operator $\mathcal{O}_{T\bar{T}}(\mu)$ modifies the field theory's UV at finite μ; the deformed CFT has a different energy spectrum and a different action. The bulk dual is AdS at finite radial cutoff r_c related to μ via a (dimension-dependent) explicit formula.
- **GeoVac $n_{\max}$ (this paper):** a *truncation of the spectrum*, not the action. The framework's discrete spectral triple at $n_{\max}$ keeps the underlying free CFT and its action intact, but restricts the mode sum to $n \le n_{\max}$. As $n_{\max} \to \infty$, the spectral data converges to the round-sphere CFT bit-exactly (Theorems 3.1, 3.2); the intermediate $n_{\max}$ values are not a TT-bar-deformed CFT but genuine spectral truncations of the original CFT.

At the partition-function-value level, the two regularizations might still be related: each produces a finite number Z_{cutoff} that converges to the continuum CFT partition function as the cutoff is removed. Whether there exists a $\mu(n_{\max}, R)$ such that $Z^{\mathrm{GeoVac}}_{n_{\max}}(R) = Z_{\text{TT-bar}}(\mu(n_{\max}, R), R)$ is an open question. A positive answer would identify the framework's spectral cutoff with a specific TT-bar-deformation strength; a negative answer would crystallize that the two cutoffs are structurally incompatible at the partition-function level.

Structural distinction from holographic correspondences (dS/CFT and AdS/CFT). The bit-exact reproduction of CFT_3-on-S^3 partition function data (Sec. 3) places the framework in structural adjacency with both AdS/CFT and dS/CFT, but in neither category. Specifically, the framework's S^3 substrate occupies the same geometric position that Euclidean dS/CFT [40] assigns to its boundary CFT_3: in Maldacena's Euclidean sharpening of Strominger's dS/CFT proposal [39], the Hartle–Hawking wavefunctional of dS_4 at I^+ equals the CFT_3-on-S^3 partition function $\Psi_{\mathrm{dS}_4}[\phi_0] = Z_{\mathrm{CFT}_3}[\phi_0]$. The KPS values reproduced bit-exactly here (Theorems 3.1, 3.2) are precisely the boundary CFT data that dS/CFT would predict via this identification.

However, the framework does *not* contain a bulk dS_4 (or its Euclidean partner S^4). The S^3 is the Fock-projection image of the Coulomb spectrum (Paper 7 [1]), not a future-infinity boundary of any 4D bulk; there is no Hartle–Hawking wavefunctional construction (no functional integral over 4D geometries); and there is no holographic dictionary mapping bulk fields to boundary operators. The CFT_3 partition function values emerge directly from spectral asymptotics of the discrete spectral triple, not from holographic reconstruction. This places the framework in a third structural category, distinct from the holographic correspondences:

Theory	Bulk	Correspondence mechanism
$\mathrm{AdS}_5/\mathrm{CFT}_4$	AdS_5 IIB SUGRA	Holographic dictionary
$\mathrm{AdS}_4/\mathrm{CFT}_3$ (ABJM)	AdS_4 M-theory	Holographic dictionary
$\mathrm{dS}_4/\mathrm{CFT}_3$	dS_4	Hartle–Hawking wavefunctional
Chern–Simons / WZW	3D topological	State-sum / topological
Connes–vS (2021) S^1	Discrete spectral triple	Propinquity convergence
GeoVac (this paper)	**Discrete spectral triple on** S^3	**State-space GH convergence at rate** $4/\pi$

The framework's analog of the holographic correspondence is the Paper 38 [8] state-space GH convergence theorem $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_\infty) \leq C_3 \gamma_{n_{\max}} \to 0$. This is a correspondence between two operator-algebraic objects (the discrete spectral triple at finite cutoff and the continuum spectral triple on round S^3), not between two physical geometries (bulk gravity and boundary CFT). The third category is structurally characterized by:

- the "bulk" replaced by a discrete spectral-triple substrate (operator-system level, finite cutoff);
- the "correspondence" replaced by state-space GH convergence in the continuum limit (Paper 38's five-lemma proof);
- boundary CFT data emerging from spectral asymptotics (zeta-function derivatives at zero), not from holographic reconstruction.

In one sentence: *the framework is the discrete spectral-triple realization of the boundary* CFT_3*-on-*S^3 *partition function data and modular structure, distinguished from AdS/CFT and dS/CFT by absence of any bulk gravitational theory and presence of a state-space-GH-convergence "correspondence" between operator-algebraic discretization levels.* This is not a new holographic correspondence (no bulk = no holography), nor a refutation of dS/CFT (the boundary data is consistent with what dS/CFT would predict); it is a structurally distinct realization that produces the same boundary CFT data via a different mechanism. The bulk-side identifications named in Prop. 6.1 for AdS_4 remain unreachable from the framework's existing infrastructure for the same reasons (no bulk geometric machinery in `geovac/`). The framework's contribution is to provide a non-holographic operationalization of the boundary-CFT-on-S^3 data, structurally distinct from both AdS/CFT and dS/CFT.

Areas yet to be explored. Beyond the immediate catalogue targets, several adjacent directions extend the framework's reach into established physics:

- **Holographic central charges** (a- and c-anomalies in even d). Henningson-Skenderis [28] and follow-ups give closed forms for the holographic dual of free CFT central charges. The framework's spectral-action machinery (Paper 28 [9]) potentially provides a discrete-cutoff reproduction; the connection to Paper 24's Bargmann-Segal S^5 is unexplored.
- **Entanglement-entropy universal terms.** The continuum Casini-Huerta-Myers [31] entanglement entropy of a ball region has a universal subleading constant equal to $-F$. Connecting this to the framework's wedge KMS structure (Sec. 4) and to the master Mellin engine decomposition would strengthen the F-as-engine-signature reading.

- **Higher-rank Lie groups beyond** SU(2)**.** The Paper 40 unified GH-convergence theorem [10] extends the state-space GH convergence machinery to all simple compact Lie groups. A natural catalogue extension is CFT-on-G partition functions for various G (heat-kernel zeta on SU(3), Sp(2), etc.) where the framework's GH-rate constant $4/\pi$ is universal but the specific transcendental ring may differ.

- **Spectral-action coefficients.** The Connes-Chamseddine [30] spectral action $\mathrm{Tr}\, f(D/\Lambda)$ has heat-kernel expansion coefficients that classify the gravitational + matter content of an almost-commutative spectral triple. These coefficients are exactly the master Mellin engine M2 sector. The catalogue could include M-engine readings of standard spectral-action calculations on S^3, S^5, and higher.

- **Lorentzian-side observables.** The catalogue here is boundary-side Riemannian. The Lorentzian extension (Papers 42-49) provides modular Hamiltonian, Pythagorean orthogonality, and twin-paradox-as-quantum-information structures. Cataloguing Lorentzian-side CFT observables (thermal partition functions on $S^3 \times \mathbb{R}_t$, Rindler-wedge entropies, etc.) is a parallel direction.

Each catalogue entry either (i) verifies the master Mellin engine on independent physics data, (ii) extends a structural cancellation theorem (like log 3 cancellation on S^5, Theorem 7.3), or (iii) identifies a wall where the simple ring fails (like the squashed-S^3 deformed-Hurwitz case, not in the MR-B Jacobi-ϑ_2 closed form). Each outcome is paper-worthy in its own right; the cumulative catalogue would constitute a structural classification of free-field CFT-on-sphere partition functions via the GeoVac spectral triple.

8.4 Place in the GeoVac math.OA series

This paper is the twelfth math.OA standalone in the GeoVac series, following Papers 38, 39, 40, 42, 43, 44, 45, 46, 47, 48, 49. The previous eleven established operator-algebraic legitimacy of the framework's discrete-spectral-triple truncation, the spectral-truncation convergence theory at various levels (Riemannian, tensor-product, universal compact-Lie-group, K^+-weak-form Lorentzian, strong-form Lorentzian, norm-resolvent non-compact, K^+-weak-form synthetic Lorentzian via Mondino–Sämann bridge, strong-form synthetic Lorentzian via OSLPLS). The present paper is the *first connecting that math.OA arc to a physics literature* — specifically the CFT-on-sphere partition function literature of [22–24, 26, 27] — with a load-bearing structural finding (the master Mellin engine verification on independent physics data) plus three bit-exact / formal correspondences (partition function, wedge boundary dimension, CHM modular Hamiltonian).

Acknowledgments

This work was developed in the GeoVac project's AI-augmented agentic workflow with Anthropic's Claude.

References

[1] J. Loutey, *Paper 7: The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology*, GeoVac internal preprint, 2026.

[2] J. Loutey, *Paper 18: Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds*, GeoVac internal preprint, 2026.

[3] J. Loutey, *Paper 22: Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems*, GeoVac internal preprint, 2026.

[4] J. Loutey, *Paper 23: Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice*, GeoVac internal preprint, 2026.

[5] J. Loutey, *Paper 29: Finite-Size Graph-RH for the GeoVac Hopf Graphs: Ihara Zeta, Bound-Crossing, and a Scope Boundary on Selberg-on-Hydrogen*, GeoVac internal preprint, 2026.

[6] J. Loutey, *Paper 32: The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac internal preprint, 2026.

[7] J. Loutey, *Paper 35: Time as Projection: The Klein-Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split*, GeoVac internal preprint, 2026.

[8] J. Loutey, *Paper 38: State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3*, GeoVac internal preprint, 2026.

[9] J. Loutey, *Paper 28: QED on S^3: Spectral Geometry of Perturbative Transcendentals*, GeoVac internal preprint, 2026.

[10] J. Loutey, *Paper 40: State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric*, GeoVac internal preprint, 2026.

[11] J. Loutey, *Paper 42: Tomita–Takesaki modular structure on truncated $SU(2)$ spectral triples: four-witness Wick-rotation literal identification at finite cutoff*, GeoVac internal preprint, 2026.

[12] J. Loutey, *Paper 43: Lorentzian extension of the four-witness Wick-rotation theorem on truncated $S^3 \times \mathbb{R}$ spectral triples at finite cutoff*, GeoVac internal preprint, 2026.

[13] J. Loutey, *Paper 44: Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, 2026.

[14] J. Loutey, *Paper 45: Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples: a degeneracy theorem*, GeoVac internal preprint, 2026.

[15] J. Loutey, *Paper 46: Strong-form Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples*, GeoVac internal preprint, 2026.

[16] J. Loutey, *Paper 47: Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on $S^3 \times \mathbb{R}_t$ and the two-rate hybrid limit*, GeoVac internal preprint, 2026.

[17] J. Loutey, *Paper 48: Krein-pointed proper quantum metric spaces and the bridge to Mondino–Sämann Lorentzian pre-length spaces, with application to the GeoVac hemispheric wedge*, GeoVac internal preprint, 2026.

[18] J. Loutey, *Paper 49: Operator-system Lorentzian pre-length spaces and the strong-form Krein–Mondino–Sämann bridge*, GeoVac internal preprint, 2026.

[19] V. Fock, *Zur Theorie des Wasserstoffatoms*, Z. Phys. **98** (1935), 145–154.

[20] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18.

[21] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067, arXiv:2004.14115.

[22] J. L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nucl. Phys. B **270** (1986), 186–204.

[23] I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *F-theorem without supersymmetry*, JHEP **10** (2011) 038, arXiv:1105.4598.

[24] D. L. Jafferis, I. R. Klebanov, S. S. Pufu, and B. R. Safdi, *Towards the F-theorem: $N = 2$ field theories on the three-sphere*, JHEP **06** (2011) 102, arXiv:1103.1181.

[25] N. Hama, K. Hosomichi, and S. Lee, *SUSY gauge theories on squashed three-spheres*, JHEP **05** (2011) 014, arXiv:1102.4716.

[26] M. Beccaria, X. Bekaert, and A. A. Tseytlin, *Partition function of free conformal higher spin theory*, JHEP **08** (2014) 113, arXiv:1406.3542.

[27] Y. Lei and S. van Leuven, *Modularity in $d > 2$ free conformal field theory*, arXiv:2406.01567, 2024.

[28] M. Henningson and K. Skenderis, *The holographic Weyl anomaly*, JHEP **07** (1998) 023, arXiv:hep-th/9806087.

[29] T. Hartman, J. Kruthoff, E. Shaghoulian, and A. Tajdini, *Holography at finite cutoff with a T^2 deformation*, JHEP **03** (2019) 004, arXiv:1807.11401.

[30] A. H. Chamseddine and A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997), 731–750.

[31] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, arXiv:1102.0440.

[32] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.

[33] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96** (2006), 181602, arXiv:hep-th/0603001.

[34] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **06** (2016) 004, arXiv:1512.06431.

[35] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, *Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence*, JHEP **06** (2015) 149, arXiv:1503.06237.

[36] GeoVac internal, *Sprint BH-Phase0 diagnostic memo*, `debug/bh_phase0_diagnostic_memo.md`, 2026-05-22.

[37] GeoVac internal, *Sprint TD Track 5 PSLQ-disjoint correlation entropy*, `debug/sprint_td_track5_memo.md`, 2026-05-09.

[38] GeoVac internal, *External-input three-class partition*, `memory/external_input_three_class_partition.m`
2026-05-25.

[39] A. Strominger, *The dS/CFT correspondence*, JHEP **10** (2001) 034, arXiv:hep-th/0106113.

[40] J. Maldacena, *Non-Gaussian features of primordial fluctuations in single field inflationary models*, JHEP **05** (2003) 013, arXiv:astro-ph/0210603.